THE NORMALIZED BLOWING UP OF A NAGATA ALGEBRAIC SPACE AT A CLOSED POINT IS A NORMAL NAGATA SPACE PROPER OVER THE BASE

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Editorial note. Assembly-type card at the H2 floor: the proof is a five-citation relay; the obligation is verifying each premise match and the composition directions.

Allowed standard prerequisites (editorial).

- Blowing up is proper over a locally Noetherian base. Divisors on Spaces, lemma blowing-up-projective (cross-chapter, named). Makes X' -> X proper.
- Finite type spaces over a Nagata base are Nagata. Morphisms of Spaces, lemma finite-type-nagata (cross-chapter, named). Shows X' is Nagata.
- Normalization of a Nagata space is finite. Morphisms of Spaces, lemma nagata-normalization (cross-chapter, named). Makes the normalization X" -> X' finite, and with the finite-proper and composition-proper lemmas X" -> X is proper.
- Normalization is normal. Morphisms of Spaces, lemma normalization-normal (cross-chapter, named). Gives normality of X".

Source reference. Original definition 0BHJ: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 648–648. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 085Z: focused-seeds/geometry/S9-spres-nagata-normalized-blowup/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04XY: focused-seeds/geometry/S9-spres-nagata-normalized-blowup/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NZ: focused-seeds/geometry/S9-spres-nagata-normalized-blowup/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0BAU: focused-seeds/geometry/S9-spres-nagata-normalized-blowup/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BB5: focused-seeds/geometry/S9-spres-nagata-normalized-blowup/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BB4: focused-seeds/geometry/S9-spres-nagata-normalized-blowup/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Definition 0BHJ if X is Nagata, then the normalized blowing up of X at x is a normal Nagata algebraic space proper over X.*

Proof. The blowup morphism $X' \to X$ is proper (as X is locally Noetherian we may apply Divisors on Spaces, Lemma 085Z). Thus X' is Nagata (Morphisms of Spaces, Lemma 0BAU). Therefore the normalization $X'' \to X'$ is finite (Morphisms of Spaces, Lemma 0BB5) and we conclude that $X'' \to X$ is proper as well (Morphisms of Spaces, Lemmas 04NZ and 04XY). It follows that the normalized blowing up is a normal (Morphisms of Spaces, Lemma 0BB4) Nagata algebraic space. □